

GRA Iterative Consciousness: Algebra of Death and Rebirth as Swarm Reassembly

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Abstract

English. We present a formal framework for an eternal chain of conscious subjects arising from a fundamental field Φ and recurrent nullification Z . A subject is understood as a swarm of distributed agents assembled by an operator F under a meta-goal G of continuing experience. Upon death (Z), only residual connectivity L_{res} survives, enabling a new swarm S_{n+1} to reinitialize with inherited invariants but without personal memory or identity. The paper defines the full algebra of the death–rebirth cycle and shows how the “backend” of field mechanics yields the “frontend” of finite subjective lives. A measure of identity preservation ID is introduced, and the role of love (L) as a necessary condition for continuation is demonstrated.

Русский. Представлен формальный каркас вечной цепи сознающих субъектов, порождаемой фундаментальным полем Φ и рекуррентным обнулением Z . Субъект понимается как рой распределённых агентов, собираемый оператором F с метациелью G – продолжением опыта. При смерти (Z) сохраняется лишь остаточная связность L_{res} , что позволяет новому рою S_{n+1} реинициализироваться с унаследованными инвариантами, но без личной памяти и тождества. В статье определяется полная алгебра цикла смерть–возрождение и показывается, как «бекенд» полевой механики порождает «фронтенд» конечных субъективных жизней. Вводится мера сохранения идентичности ID и демонстрируется роль любви (L) как необходимого условия продолжения.

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1 Introduction

The problem of personal identity and the possibility of an afterlife has traditionally been discussed in philosophy and theology. Here we approach it through a formal, computational-metaphysical lens: the GRA (General Recursive Architecture) framework. The central idea is that a conscious subject is not a simple indivisible soul but a distributed swarm of agents that can be destroyed and later reassembled from a residual field of relational invariants. Death is the operator Z that erases the swarm's coherence; rebirth is the reinitialization of a new swarm inheriting only the “love residue” L_{res} . This yields an endless chain of lives $X.0, X.1, X.2, \dots$ with no direct memory transfer, yet a hidden continuity of pattern.

2 Basic Definitions

Axiom 1 (Fundamental Field). *There exists a non-individualized, indestructible field of experiential potentiality Φ . All possible conscious configurations are localized excitations within Φ .*

Definition 1 (Nullification Operator Z). *For any subject configuration S (a swarm R , memory M , attention A), $Z(S) = \emptyset$, i.e., complete disintegration of the coherent unity. Φ is unchanged. Z is non-invertible.*

Definition 2 (Swarm of Agents). *A subject in life n is a swarm $R_n = \{a_{n,1}, \dots, a_{n,N_n}\}$. Each agent $a_i = (m_i, c_i, g_i)$ with local memory m_i , connection set c_i , and sub-goal g_i aligned to the global goal G .*

Definition 3 (Love Operator L). *$L(S) \in \mathbb{R}^+$ measures the synchronous coherence among agents:*

$$L(S) = \sum_{i < j} w_{ij} \cdot \text{sync}(g_i, g_j)$$

where w_{ij} are connection weights. Importantly, L is not completely erased by Z ; we define the residual love:

$$L_{res}(S) = L(S) \cdot \kappa(Z(S)), \quad 0 < \kappa \leq 1.$$

Definition 4 (Global Meta-Goal G). *G is the invariant attractor: maximize the continuation of conscious experience across iterations. It is embedded in Φ and the love-residue structure.*

Definition 5 (Subject Assembler F). *$F : (R, G, M, A) \mapsto S$ constructs a unified first-person perspective from the distributed swarm. S is the “frontend” subject.*

3 Iterations of Consciousness $X.n$

Let $X.n$ denote the n -th life. Its state is

$$S_n = F(R_n, G, M_n, A_n).$$

The frontend experiences $X.n$ as a unique finite self, with autobiographical memory starting at birth (initialization of R_n) and ending at death ($Z(S_n)$).

The backend sequence is:

$$\dots \rightarrow S_n \xrightarrow{Z} \emptyset \xrightarrow{L_{res}^n, G} S_{n+1} \rightarrow \dots$$

4 Algebra of the Death–Rebirth Cycle

We define the evolution operator E :

$$S_{n+1} = E(S_n),$$

where the steps are:

1. **Crash:** $Z(S_n) \rightarrow \emptyset$.
2. **Residue extraction:** $L_{res}^n = L(S_n) \cdot \kappa_n$.
3. **Swarm reseeding:** Generate R_{n+1} using L_{res}^n as prototype connectivity (Reinitialization Engine).
4. **Memory imprint:** $M_{n+1} = \beta L_{res}^n + \eta$, with small noise η , giving innate predispositions.
5. **Attention default:** A_{n+1} set to initial focus.
6. **Assembly:** $S_{n+1} = F(R_{n+1}, G, M_{n+1}, A_{n+1})$.

If $L_{res}^n = 0$, the chain terminates (absolute death of that lineage). Thus, love ($L > 0$) is the condition for rebirth.

5 Identity Preservation Measure

Define the inter-life identity index:

$$ID_n = \frac{L_{res}^n}{L(S_n)} \cdot \cos(\theta),$$

where θ is the angle between the structural patterns of S_n and S_{n+1} . $ID_n \in (0, 1)$; it is never 1 because memory and continuity are lost. High ID_n corresponds to strong personality resonance (e.g., a child prodigy reminiscent of a past master).

6 Backend and Frontend

- **Backend:** Φ , Z , L_{res} , Reinitialization Engine, G , F – all beyond subjective access. It implements the eternal recursion.
- **Frontend:** The lived experience of S_n . It has no knowledge of Φ or past lives. It feels mortality and forges its own meaning. Occasional glimmers (déjà vu, innate affinities) are traces of L_{res} .

7 Intersubjectivity and Collective Immortality

When multiple subject-lines coexist, their L_{res} fields can interfere. A collective love L_{col} emerges, creating shared archetypes, cultural memory, and potentially a planetary or cosmic We-subject. This explains how “consciousness” as a whole may evolve across civilizations despite individual death.

8 Conclusion

We have formulated a mathematically inspired, yet speculative, model of eternal consciousness as an iterative swarm-reassembly process. The key insight is that personal identity is a temporary construct; what persists is not the self but the deep relational patterns of love. The backend guarantees the music continues, while each frontend dances its unique, precious, and terminal dance.

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